



2

Mechanics

2.1 Kinematics

Assessment statements

- 2.1.1 Define displacement, velocity, speed and acceleration.
- 2.1.2 Explain the difference between instantaneous and average values of speed, velocity and acceleration.
- 2.1.3 Outline the conditions under which the equations for uniformly accelerated motion may be applied.
- 2.1.9 Determine relative velocity in one and in two dimensions.

In Chapter 1, we observed that things move and now we are going to mathematically model that movement. Before we do that, we must define some quantities that we are going to use.

Displacement and distance

It is important to understand the difference between distance travelled and displacement. To explain this, consider the route marked out on the map shown in Figure 2.1

Displacement is the distance moved in a particular direction.

The unit of displacement is the metre (m).

Displacement is a vector quantity.

On the map, the displacement is the length of the straight line from A to B, a distance of 5 km west. (Note: since displacement is a vector you should always say what the direction is.)

Distance is how far you have travelled from A to B.

The unit of distance is also the metre.

Distance is a scalar quantity.

In this example, the distance travelled is the length of the path taken, which is about 10 km.

Sometimes this difference leads to a surprising result. For example, if you run all the way round a running track you will have travelled a distance of 400 m but your displacement will be 0 m.

In everyday life, it is often more important to know the distance travelled. For example, if you are going to travel from Paris to Lyon by road you will want to know that the distance by road is 450 km, not that your final displacement will be

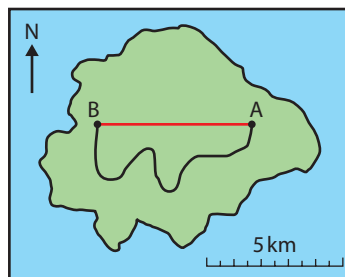


Figure 2.1

336 km SE. However, in physics, we break everything down into its simplest units, so we start by considering motion in a straight line only. In this case it is more useful to know the displacement, since that also has information about which direction you have moved.

Velocity and speed

Both speed and velocity are a measure of how fast a body is moving, but velocity is a vector quantity and speed is a scalar.

Velocity is defined as the displacement per unit time.

$$\text{velocity} = \frac{\text{displacement}}{\text{time}}$$

The unit of velocity is m s^{-1} .

Velocity is a vector quantity.

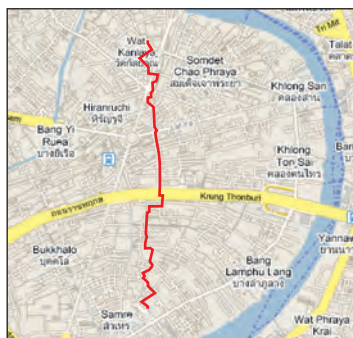
Speed is defined as the distance travelled per unit time.

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

The unit of speed is also m s^{-1} .

Speed is a scalar quantity.

Figure 2.2 It's not possible to take this route across Bangkok with a constant velocity.



The bus in the photo has a constant velocity for a very short time.



Exercise

- 1 Convert the following speeds into m s^{-1} .
 - (a) A car travelling at 100 km h^{-1}
 - (b) A runner running at 20 km h^{-1}

Average velocity and instantaneous velocity

Consider travelling by car from the north of Bangkok to the south – a distance of about 16 km. If the journey takes 4 hours, you can calculate your velocity to be $\frac{16}{4} = 4 \text{ km h}^{-1}$ in a southwards direction. This doesn't tell you anything about the journey, just the difference between the beginning and the end (unless you managed to travel at a constant speed in a straight line). The value calculated is the **average velocity** and in this example it is quite useless. If we broke the trip down into lots of small pieces, each lasting only one second, then for each second the car could be considered to be travelling in a straight line at a constant speed. For these short stages we could quote the car's **instantaneous velocity** – that's how fast it's going at that moment in time and in which direction.

Measuring velocity

You can measure velocity with a photogate connected to a timer or computer. When a card passes through the gate it is sensed by the timer, switching it on or off.

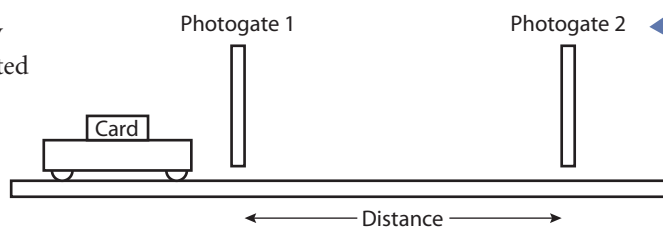


Figure 2.3 Experimental set up for measuring velocity.

$$\text{average velocity} = \frac{\text{distance}}{\text{time taken to travel between photogates}}$$

$$\text{instantaneous velocity} = \frac{\text{length of card}}{\text{time for card to pass through gate}}$$

Velocity is relative

When quoting the velocity of a body, it is important to say what the velocity is measured relative to. Consider the people in Figure 2.4

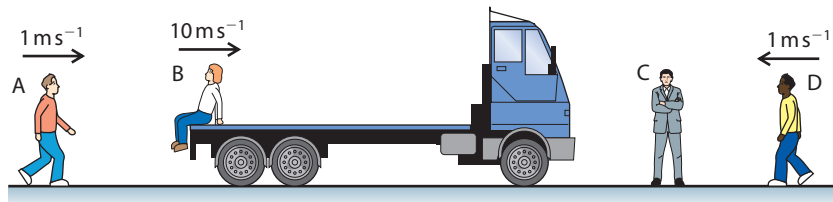


Figure 2.4 Two observers measuring the same velocity.

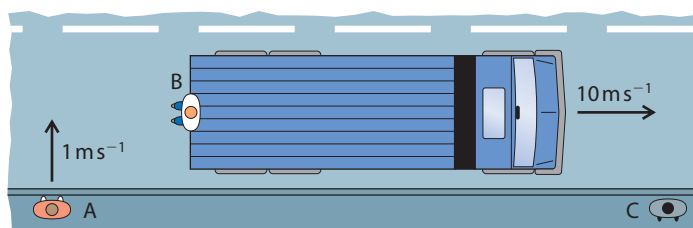
C measures the velocity of A to be 1 m s^{-1} but to B (moving on the truck towards C) the velocity of A is -9 m s^{-1} (B will see A moving away in a negative direction). You might think that A can't have two velocities, but he can – velocity is relative. In this example there are two observers, B and C. Each observer has a different 'frame of reference'. To convert a velocity, to B's frame of reference, we must subtract the velocity of B relative to C; this is 10 m s^{-1} .

So the velocity of A relative to B = $1 - 10 = -9 \text{ m s}^{-1}$

We can try the same with D who has a velocity of -1 m s^{-1} measured by C and $-1 - 10 = -11 \text{ m s}^{-1}$ measured by B.

This also works in two dimensions as follows:

A now walks across the road as illustrated by the aerial view in Figure 2.5. The velocity of A relative to C is 1 m s^{-1} north.



The velocity of A relative to B can now be found by subtracting the vectors as shown in Figure 2.6.

Examiner's hint: Velocity is a vector. If the motion is in one dimension, the direction of velocity is given by its sign. Generally, right is positive and left negative.

Figure 2.5 A now walks across the road..

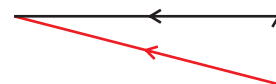


Figure 2.6 Subtracting vectors gives the relative velocity.

Exercise

- An observer standing on a road watches a bird flying east at a velocity of 10 m s^{-1} . A second observer, driving a car along the road northwards at 20 m s^{-1} sees the bird. What is the velocity of the bird relative to the driver?

Acceleration

In everyday usage, the word *accelerate* means to go faster. However in physics: acceleration is defined as the rate of change of velocity.

$$\text{acceleration} = \frac{\text{change of velocity}}{\text{time}}$$

The unit of acceleration is m s^{-2} .

Acceleration is a vector quantity.

This means that whenever a body changes its velocity it accelerates. This could be because it is getting faster, slower or just changing direction. In the example of the journey across Bangkok, the car would have been slowing down, speeding up and going round corners almost the whole time, so it would have had many different accelerations. However, this example is far too complicated for us to consider in this course (and probably any physics course). For most of this chapter we will only consider the simplest example of accelerated motion, constant acceleration.

Bodies

When we refer to a body in physics we generally mean a ball not a human body.



Constant acceleration in one dimension

In one-dimensional motion, the acceleration, velocity and displacement are all in the same direction. This means they can simply be added without having to draw triangles. Figure 2.7 shows a body that is starting from an initial velocity u and accelerating at a constant rate to velocity v in t seconds. The distance travelled in this time is s . Since the motion is in a straight line, this is also the displacement.

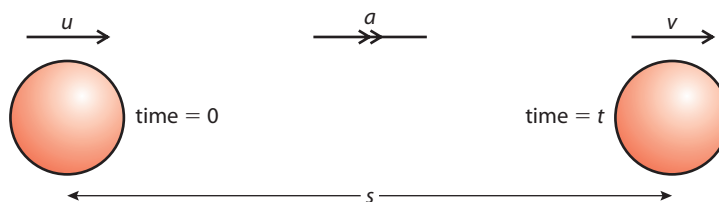


Figure 2.7 A red ball is accelerated at a constant rate.

Using the definitions already stated, we can write equations related to this example.

Average velocity

From the definition, the average velocity = $\frac{\text{displacement}}{\text{time}}$

$$\text{So average velocity} = \frac{s}{t} \quad (1)$$

Since the velocity changes at a constant rate from the beginning to the end, we can also calculate the average velocity by adding the velocities and dividing by two.

$$\text{Average velocity} = \frac{(u + v)}{2} \quad (2)$$

Acceleration

Acceleration is defined as the rate of change of velocity.

$$\text{So } a = \frac{(v - u)}{t} \quad (3)$$

We can use these equations to solve any problem involving constant acceleration. However, to make problem solving easier, we can derive two more equations by substituting from one into the other.

Equating equations (1) and (2)

$$\frac{s}{t} = \frac{(u + v)}{2}$$

$$\text{so } s = \frac{(u + v) t}{2} \quad (4)$$

Rearranging (3) gives $v = u + at$

If we substitute for v in equation (4) we get $s = ut + \frac{1}{2}at^2$ (5)

Rearranging (3) again gives $t = \frac{(v - u)}{a}$

If t is now substituted in equation (4) we get $v^2 = u^2 + 2as$ (6)

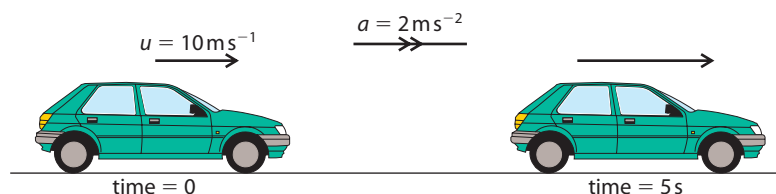
These equations are sometimes known as the *suvat* equations. If you know any 3 of *suva* and t you can find either of the other two in one step.

Worked example

1 A car travelling at 10 m s^{-1} accelerates at 2 m s^{-2} for 5 s. What is its displacement?

Solution

The first thing to do is draw a simple diagram like Figure 2.8.



This enables you to see what is happening at a glance rather than reading the text. The next stage is to make a list of *suvat*.

$$\begin{aligned} s &= ? \\ u &= 10 \text{ m s}^{-1} \\ v &= ? \\ a &= 2 \text{ m s}^{-2} \\ t &= 5 \text{ s} \end{aligned}$$

To find s you need an equation that contains *suat*. The only equation with all 4 of these quantities is $s = ut + \frac{1}{2}at^2$

Using this equation gives:

$$\begin{aligned} s &= 10 \times 5 + \frac{1}{2} \times 2 \times 5^2 \\ s &= 75 \text{ m} \end{aligned}$$

The sign of displacement, velocity and acceleration

We must not forget that displacement, velocity and acceleration are vectors. This means that they have direction. However, since this is a one-dimensional example, there are only two possible directions, forward and backward. We know which direction the quantity is in from the sign.

A positive displacement means that the body has moved right.

A positive velocity means the body is moving to the right.

suvat equations

$$\begin{aligned} a &= \frac{(v - u)}{t} \\ s &= \frac{(u + v) t}{2} \\ s &= ut + \frac{1}{2}at^2 \\ v^2 &= u^2 + 2as \end{aligned}$$

Figure 2.8 A simple diagram is always the best start.

● **Examiner's hint:** You don't need to include units in all stages of a calculation, just the answer.

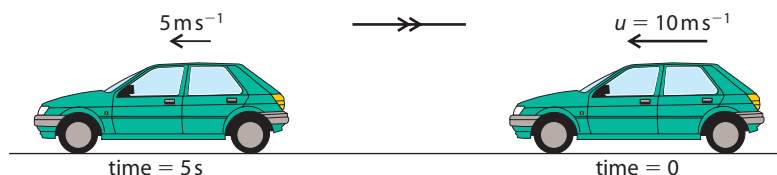


Figure 2.9

A positive acceleration means that the body is either moving to the right and getting faster or moving to the left and getting slower. This can be confusing, so consider the following example.

The car is travelling in a negative direction so the velocities are negative.

$$u = -10 \text{ m s}^{-1}$$

$$v = -5 \text{ m s}^{-1}$$

$$t = 5 \text{ s}$$

The acceleration is therefore given by

$$a = \frac{(v - u)}{t} = \frac{-5 - (-10)}{5} = 1 \text{ m s}^{-2}$$

The positive sign tells us that the acceleration is in a positive direction (right) even though the car is travelling in a negative direction (left).

Example

A body with a constant acceleration of -5 m s^{-2} is travelling to the right with a velocity of 20 m s^{-1} . What will its displacement be after 20 s?

$$s = ?$$

$$u = 20 \text{ m s}^{-1}$$

$$v = ?$$

$$a = -5 \text{ m s}^{-2}$$

$$t = 20 \text{ s}$$

To calculate s we can use the equation $s = ut + \frac{1}{2}at^2$

$$s = 20 \times 20 + \frac{1}{2}(-5) \times 20^2 = 400 - 1000 = -600 \text{ m}$$

This means that the final displacement of the body is to the left of the starting point. It has gone forward, stopped and then gone backwards.

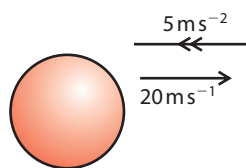


Figure 2.10 The acceleration is negative so pointing to the left.

g

The acceleration due to gravity is not constant all over the Earth. 9.81 m s^{-2} is the average value. The acceleration also gets smaller the higher you go. However we ignore this change when conducting experiments in the lab since labs aren't that high.

To make the examples easier to follow, $g = 10 \text{ m s}^{-2}$ is used throughout; you should only use this approximate value in exam questions if told to do so.

Exercises

- Calculate the final velocity of a body that starts from rest and accelerates at 5 m s^{-2} for a distance of 100 m.
- A body starts with a velocity of 20 m s^{-1} and accelerates for 200 m with an acceleration of 5 m s^{-2} . What is the final velocity of the body?
- A body accelerates at 10 m s^{-2} reaching a final velocity of 20 m s^{-1} in 5 s. What was the initial velocity of the body?

2.2 Free fall motion

Assessment statements

- Identify the acceleration of a body falling in a vacuum near the Earth's surface with the acceleration g of free fall.
- Solve problems involving the equations of uniformly accelerated motion.
- Describe the effects of air resistance on falling objects.



Although a car was used in one of the previous illustrations, the acceleration of a car is not usually constant, so we shouldn't use the *suvat* equations. The only example of constant acceleration that we see in everyday life is when a body is dropped. Even then the acceleration is only constant for a short distance.

Acceleration of free fall

When a body is allowed to fall freely we say it is in free fall. Bodies falling freely on the Earth fall with an acceleration of about 9.81 m s^{-2} . (It depends where you are.) The body falls because of gravity. For that reason we use the letter *g* to denote this acceleration. Since the acceleration is constant, we can use the *suvat* equations to solve problems.

Exercises

In these calculations use $g = 10 \text{ m s}^{-2}$.

- 6 A ball is thrown upwards with a velocity of 30 m s^{-1} . What is the displacement of the ball after 2 s?
- 7 A ball is dropped. What will its velocity be after falling 65 cm?
- 8 A ball is thrown upwards with a velocity of 20 m s^{-1} . After how many seconds will the ball return to its starting point?



The effect of air resistance

If you jump out of a plane (with a parachute on) you will feel the push of the air as it rushes past you. As you fall faster and faster, the air will push upwards more and more until you can't go any faster. At this point you have reached terminal velocity. We will come back to this example after introducing forces.

Measuring *g*

Measuring *g* by timing a ball falling from different heights is a common physics experiment that you could well perform in the practical programme of the IB course. There are various different ways of doing this but a common method is to use a timer that starts when the ball is released and stops when it hits a platform. An example of this apparatus is shown in the photo. The distance travelled by the ball and the time taken are related by the *suvat* equation $s = ut + \frac{1}{2}at^2$. This simplifies to $s = \frac{1}{2}at^2$ since the initial velocity is zero. This means that *s* is proportional to t^2 so if you plot a graph of *s* against t^2 you will get a straight line whose gradient is $\frac{1}{2}g$.



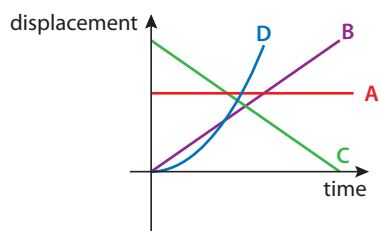
Free fall apparatus.

2.3 Graphical representation of motion

Assessment statements

- 2.1.7 Draw and analyse distance–time graphs, displacement–time graphs, velocity–time graphs and acceleration–time graphs.
- 2.1.8 Calculate and interpret the gradients of displacement–time graphs and velocity–time graphs, and the areas under velocity–time graphs and acceleration–time graphs.

Figure 2.11 Graphical representation of motion.



Line A

A body that is not moving.

Displacement is always the same.

Velocity is zero.

Acceleration is zero.

Line B

A body that is travelling with a constant positive velocity.

Displacement increases linearly with time.

Velocity is a constant positive value.

Acceleration is zero.

Line C

A body that has a constant negative velocity.

Displacement is decreasing linearly with time.

Velocity is a constant negative value.

Acceleration is zero.

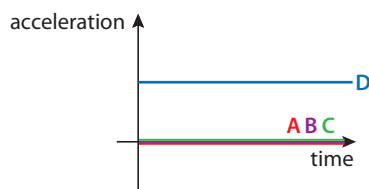
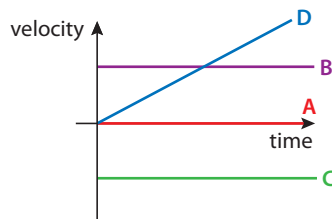
Line D

A body that is accelerating with constant acceleration.

Displacement is increasing at a non-linear rate. The shape of this line is a parabola since displacement is proportional to t^2 ($s = ut + \frac{1}{2}at^2$).

Velocity is increasing linearly with time.

Acceleration is a constant positive value.



● **Examiner's hint:** You need to be able to

- figure out what kind of motion a body has by looking at the graphs
- sketch graphs for a given motion.

The best way to go about sketching graphs is to split the motion into sections then plot where the body is at different times; joining these points will give the displacement–time graph. Once you have done that you can work out the v – t and a – t graphs by looking at the s – t graph rather than the motion.

Gradient of displacement–time

The gradient of a graph is $\frac{\text{change in } y}{\text{change in } x}$

$$= \frac{\Delta y}{\Delta x}$$

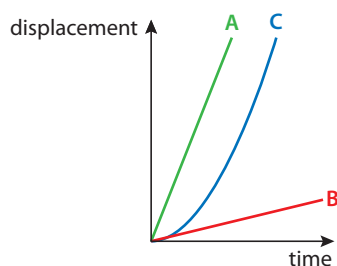
In the case of the displacement–time graph this will give

$$\text{gradient} = \frac{\Delta s}{\Delta t}$$

This is the same as velocity.

So the gradient of the displacement–time graph equals the velocity. Using this information, we can see that line A in Figure 2.12 represents a body with greater velocity than line B and that since the gradient of line C is increasing, this must be the graph for an accelerating body.

Figure 2.12



Instantaneous velocity

When a body accelerates its velocity is constantly changing. The displacement–time graph for this motion is therefore a curve. To find the instantaneous velocity from the graph we can draw a tangent to the curve and find the gradient of the tangent as shown in Figure 2.13.

Area under velocity–time graph

The area under the velocity–time graph for the body travelling at constant velocity v shown in Figure 2.14 is given by

$$\text{area} = v\Delta t$$

But we know from the definition of velocity that $v = \frac{\Delta s}{\Delta t}$

Rearranging gives $\Delta s = v\Delta t$ so the area under a velocity–time graph gives the displacement.

This is true not only for simple cases such as this but for all examples.

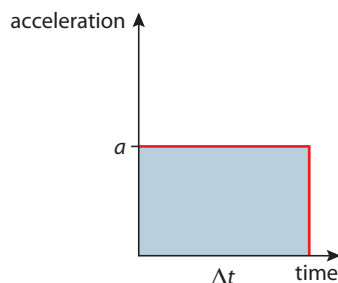
Gradient of velocity–time graph

The gradient of the velocity–time graph is given by $\frac{\Delta v}{\Delta t}$. This is the same as acceleration.

Area under acceleration–time graph

The area under an acceleration–time graph in Figure 2.15 is given by $a\Delta t$. But we know from the definition of acceleration that $a = \frac{(v - u)}{t}$

Rearranging this gives $v - u = a\Delta t$ so the area under the graph gives the change in velocity.



If you have covered calculus in your maths course you may recognise these equations:

$$v = \frac{ds}{dt} \Rightarrow s = \int v dt, \quad a = \frac{dv}{dt} \Rightarrow v = \int a dt$$

Exercises

- 9 Sketch a velocity–time graph for a body starting from rest and accelerating at a constant rate to a final velocity of 25 ms^{-1} in 10 seconds. Use the graph to find the distance travelled and the acceleration of the body.
- 10 Describe the motion of the body whose velocity–time graph is shown in Figure 2.16. What is the final displacement of the body?

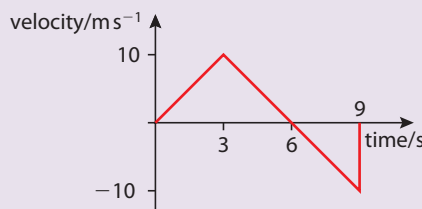


Figure 2.16

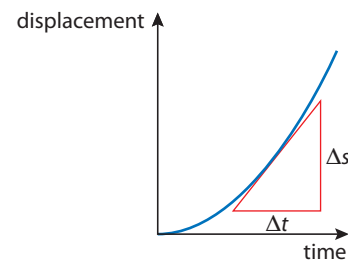


Figure 2.13

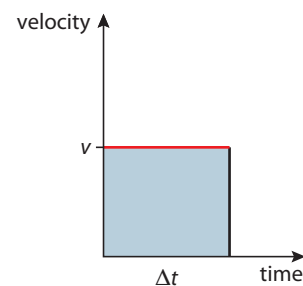


Figure 2.14

Figure 2.15

Example 1: the *suvat* example

As an example let us consider the motion we looked at when deriving the *suvat* equations.

Figure 2.17 A body with constant acceleration.

Negative time

Negative time doesn't mean going back in time – it means the time before you started the clock.

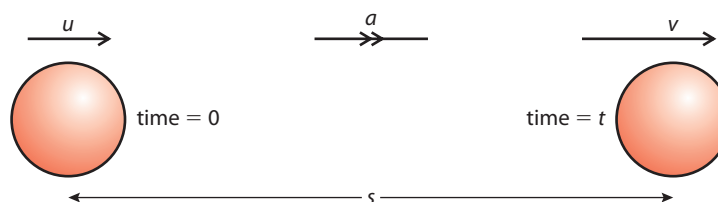
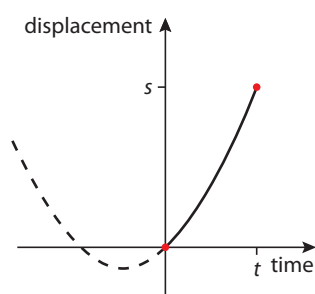


Figure 2.18



Displacement-time

The body starts with velocity u and travels to the right with constant acceleration, a for a time t . If we take the starting point to be zero displacement, then the displacement-time graph starts from zero and rises to s in t seconds. We can therefore plot the two points shown in Figure 2.18. The body is accelerating so the line joining these points is a parabola. The whole parabola has been drawn to show what it would look like – the reason it is offset is because the body is not starting from rest. The part of the curve to the left of the origin tells us what the particle was doing before we started the clock.

Velocity-time

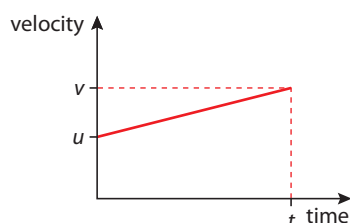
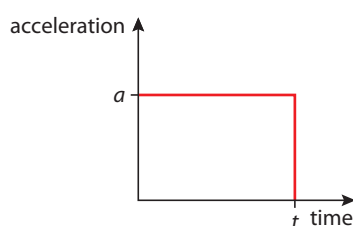


Figure 2.19 is a straight line with a positive gradient showing that the acceleration is constant. The line doesn't start from the origin since the initial velocity is u . The gradient of this line is $\frac{(v-u)}{t}$ which we know from the *suvat* equations is acceleration.

The area under the line makes the shape of a trapezium. The area of this trapezium is $\frac{1}{2}(v+u)t$. This is the *suvat* equation for s .

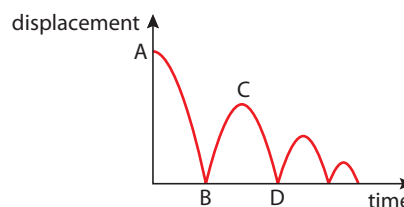
Figure 2.20



Acceleration-time

The acceleration is constant so the acceleration-time graph is simply a horizontal line as shown in Figure 2.20. The area under this line is $a \times t$ which we know from the *suvat* equations equals $(v-u)$.

Example 2: The bouncing ball



Consider a rubber ball dropped from some position above the ground A onto a hard surface B. The ball bounces up and down several times. Figure 2.21 shows the

displacement–time graph for 4 bounces. From the graph we see that the ball starts above the ground then falls with increasing velocity (as deduced by the increasing negative gradient). When the ball bounces at B the velocity suddenly changes from negative to positive as the ball begins to travel back up. As the ball goes up, its velocity gets less until it stops at C and begins to fall again.

Exercise

- 11** By considering the gradient of the displacement–time graph in Figure 2.21 plot the velocity–time graph for the motion of the bouncing ball.



To view a simulation that enables you to plot the graphs as you watch the motion, visit www.heinemann.co.uk/hotlinks, enter the express code 4426P and click on Weblink 2.1.

Example 3: A ball falling with air resistance

Figure 2.22 represents the motion of a ball that is dropped several hundred metres through the air. It starts from rest and accelerates for some time. As the ball accelerates, the air resistance gets bigger, which prevents the ball from getting any faster. At this point the ball continues with constant velocity.

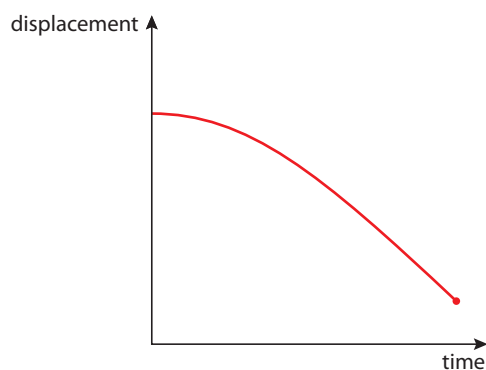


Figure 2.22

Exercise

- 12** By considering the gradient of the displacement–time graph plot the velocity–time graph for the motion of the falling ball.

2.4 Projectile motion

Assessment statements

- 9.1.1 State the independence of the vertical and the horizontal components of velocity for a projectile in a uniform field.
- 9.1.2 Describe and sketch the trajectory of projectile motion as parabolic in the absence of air resistance.
- 9.1.3 Describe qualitatively the effect of air resistance on the trajectory of a projectile.
- 9.1.4 Solve problems on projectile motion.

A stroboscopic photograph of a projected ball.

We all know what happens when a ball is thrown; it follows a curved path like the one in the photo below. We can see from this photo that the path is parabolic, and later we will show why that is the case.

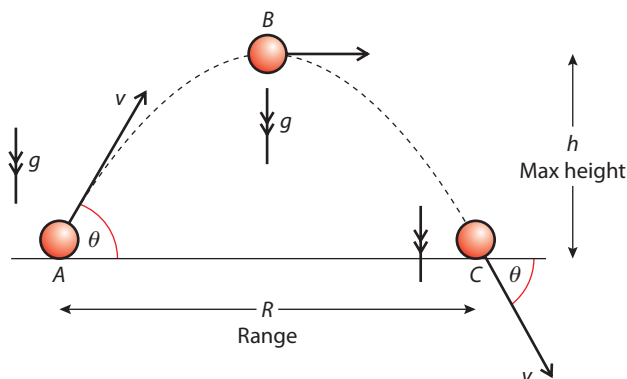


Modelling projectile motion

All examples of motion up to this point have been in one dimension but projectile motion is two-dimensional. However, if we take components of all the vectors vertically and horizontally, we can simplify this into two simultaneous one-dimensional problems. The important thing to realise is that the vertical and horizontal components are independent of each other; you can test this by dropping a stone off a cliff and throwing one forward at the same time, they both hit the bottom together. The downward motion is not altered by the fact that one is also moving forward.

Consider a ball that is projected at an angle θ to the horizontal, as shown in Figure 2.23. We can split the motion into three parts, beginning, middle and end, and analyse the vectors representing displacement, velocity and time at each stage. Note that since the path is symmetrical, the motion on the way down is the same as the way up.

Figure 2.23 A projectile launched at an angle θ .



Horizontal components

At A (time = 0)	At B (time = $\frac{t}{2}$)	At C (time = t)
Displacement = zero	Displacement = $\frac{R}{2}$	Displacement = R
Velocity = $v \cos \theta$	Velocity = $v \cos \theta$	Velocity = $v \cos \theta$
Acceleration = 0	Acceleration = 0	Acceleration = 0

Vertical components

At A	At B	At C
Displacement = zero	Displacement = h	Displacement = zero
Velocity = $v \sin \theta$	Velocity = zero	Velocity = $-v \sin \theta$
Acceleration = $-g$	Acceleration = $-g$	Acceleration = $-g$

We can see that the vertical motion is constant acceleration and the horizontal motion is constant velocity. We can therefore use the *suvat* equations.

suvat for horizontal motion

Since acceleration is zero there is only one equation needed to define the motion

<i>suvat</i>	A to C
Velocity = $v = \frac{s}{t}$	$R = v \cos \theta t$

suvat for vertical motion

When considering the vertical motion it is worth splitting the motion into two parts.

<i>suvat</i>	At B	At C
$s = \frac{1}{2}(u + v)t$	$h = \frac{1}{2}(v \sin \theta) \frac{t}{2}$	$0 = \frac{1}{2}(v \sin \theta - v \sin \theta)t$
$v^2 = u^2 + 2as$	$0 = v^2 \sin^2 \theta - 2gh$	$(-v \sin \theta)^2 = (v \sin \theta)^2 - 0$
$s = ut + \frac{1}{2}at^2$	$h = v \sin \theta t - \frac{1}{2}g\left(\frac{t}{2}\right)^2$	$0 = v \sin \theta t - \frac{1}{2}gt^2$
$a = \frac{v - u}{t}$	$g = \frac{v \sin \theta - 0}{\frac{t}{2}}$	$g = \frac{v \sin \theta - -v \sin \theta}{t}$

Some of these equations are not very useful since they simply state that $0 = 0$. However we do end up with three useful ones (highlighted):

$$R = v \cos \theta t \quad (1)$$

$$0 = v^2 \sin^2 \theta - 2gh \quad \text{or} \quad h = \frac{v^2 \sin^2 \theta}{2g} \quad (2)$$

$$0 = v \sin \theta t - \frac{1}{2}gt^2 \quad \text{or} \quad t = \frac{2v \sin \theta}{g} \quad (3)$$

Solving problems

In a typical problem you will be given the magnitude and direction of the initial velocity and asked to find either the maximum height or range. To calculate h you can use equation (2) but to calculate R you need to find the time of flight so must use (3) first (you could also substitute for t into equation (1) to give a fourth equation but maybe we have enough equations already).

You do not have to remember a lot of equations to solve a projectile problem. If you understand how to apply the *suvat* equations to the two components of the projectile motion, you only have to remember the *suvat* equations (and they are in the databook).

Parabolic path

Since the horizontal displacement is proportional to t the path has the same shape as a graph of vertical displacement plotted against time. This is parabolic since the vertical displacement is proportional to t^2 .

Maximum range

For a given value of v the maximum range is when $v \cos \theta t$ is a maximum value. Now $t = \frac{2v \sin \theta}{g}$. If we substitute this for t we get

$$R = \frac{2v^2 \cos \theta \sin \theta}{g}$$

This is a maximum when $\cos \theta \sin \theta$ is maximum, which is when $\theta = 45^\circ$.

To view a simulation of projectile motion, visit www.heinemann.co.uk/hotlinks, enter the express code 4426P and click on Weblink 2.2.

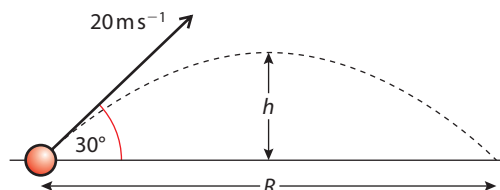


Worked example

- 1 A ball is thrown at an angle of 30° to the horizontal at a speed of 20 m s^{-1} . Calculate its range and the maximum height reached.
- 2 A ball is thrown horizontally from a cliff top with a horizontal speed of 10 m s^{-1} . If the cliff is 20 m high what is the range of the ball?

Solution

- 1 First, as always, draw a diagram, including labels defining all the quantities known and unknown.



Now we need to find the time of flight. If we apply $s = ut + \frac{1}{2}at^2$ to the whole flight we get

$$t = \frac{2v \sin \theta}{g} = \frac{(2 \times 20 \times \sin 30^\circ)}{10} = 2 \text{ s}$$

We can now apply $s = vt$ to the whole flight to find the range:

$$R = v \cos \theta t = 20 \times \cos 30^\circ \times 2 = \mathbf{34.6 \text{ m}}$$

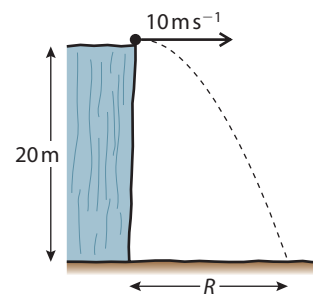
Finally to find the height, we use $s = ut + \frac{1}{2}at^2$ to the vertical motion, but remember, this is only half the complete flight so the time is 1 s .

$$h = v \sin \theta t - \frac{1}{2}gt^2 = 20 \times \sin 30^\circ \times 1 - \frac{1}{2} \times 10 \times 1^2 = 10 - 5 = \mathbf{5 \text{ m}}$$

- 2 This is an easy one since there aren't any angles to deal with. The initial vertical component of the velocity is zero and the horizontal component is 10 m s^{-1} . To calculate the time of flight we apply $s = ut + \frac{1}{2}at^2$ to the vertical component. Knowing that the final displacement is -20 m this gives

$$-20 \text{ m} = 0 - \frac{1}{2}gt^2 \text{ so } t = \sqrt{\frac{(2 \times 20)}{10}} = 2 \text{ s}$$

We can now use this value to find the range by applying the formula $s = vt$ to the horizontal component: $R = 10 \times 2 = \mathbf{20 \text{ m}}$



If you have ever played golf you will know it is not true that the maximum range is achieved with an angle of 45° , it's actually much less. This is because the ball is held up by the air like an aeroplane is. In this photo Alan Shepard is playing golf on the moon. Here the maximum range will be at 45° .

Exercises

- 13 Calculate the range of a projectile thrown at an angle of 60° to the horizontal with velocity 30 m s^{-1} .
- 14 You throw a ball at a speed of 20 m s^{-1} .
 - (a) At what angle must you throw it so that it will just get over a wall that is 5 m high?
 - (b) How far away from the wall must you be standing?
- 15 A gun is aimed so that it points directly at the centre of a target 200 m away. If the bullet travels at 200 m s^{-1} how far below the centre of the target will the bullet hit?
- 16 If you can throw a ball at 20 m s^{-1} what is the maximum distance you can throw it?

Projectile motion with air resistance

In all the examples above we have ignored the fact that the air will resist the motion of the ball. The actual path of a ball including air resistance is likely to be as shown in Figure 2.24.

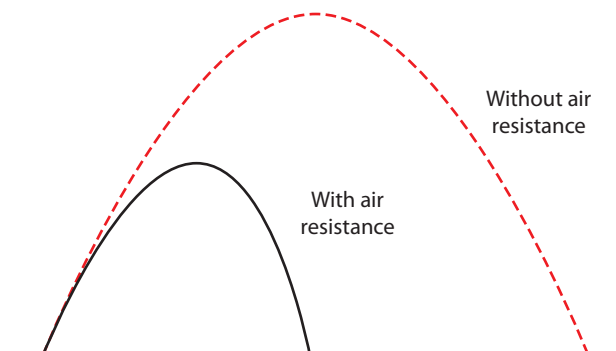


Figure 2.24

Notice both the height and range are less. It is also no longer a parabola – the way down is steeper than the way up.

2.5 Forces and dynamics

Assessment statements

- 2.2.1 Calculate the weight of a body using the expression $W = mg$.
- 2.2.2 Identify the forces acting on an object and draw free body diagrams representing the forces acting.
- 2.2.3 Determine the resultant force in different situations.

Forces

From experience, we know that things don't seem to move unless we push them, so movement is related to pushing. In this next section we will investigate this relationship.

What is a force?

A force is simply a push or a pull.

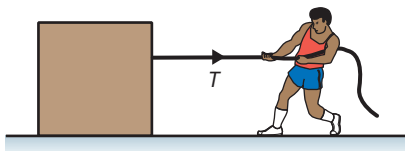
The unit of force is the newton (N).

Force is a vector quantity.

You might believe that there are hundreds of different ways to push or pull an object but there are actually surprisingly few.

1. Tension

If you attach a rope to a body and pull it, the rope is in tension. This is also the name of the force exerted on the body.



The size of one newton

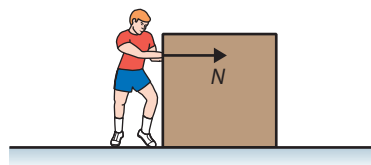
If you hold an object of mass 100 g in your hand then you will be exerting an upward force of about one newton (1 N).

Figure 2.25 The force experienced by the block is tension, T .

2. Normal force

Whenever two surfaces are in contact, there will be a force between them (if not then they are not in contact). This force acts at right angles to the surface so is called the *normal force*.

Figure 2.26 The man pushes the block with his hands. The force is called the normal force, N .



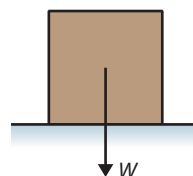
3. Gravitational force

We know that all objects experience a force that pulls them downwards; we call this force the *weight*. The direction of this force is always towards the centre of the Earth. The weight of a body is directly proportional to the mass of the body.

$W = mg$ where g = the acceleration of free fall.

You will discover why this is the case later in the chapter.

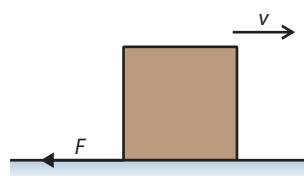
Figure 2.27 This box is pulled downwards by gravity. We call this force the weight, W .



Where to draw forces

It is important that you draw the point of application of the forces in the correct place. Notice where the forces are applied in these diagrams.

Figure 2.28 This box sliding along the floor will slow down due to the friction, F , between it and the floor.



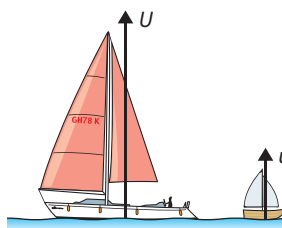
4. Friction force

Whenever two touching surfaces move, or attempt to move, relative to each other, there is a force that opposes the motion. This is called *frictional force*. The size of this force is dependent on the material of the surfaces and how much force is used to push them together.

5. Upthrust

Upthrust is the name of the force experienced by a body immersed in a fluid (gas or liquid). This is the force that pushes up on a boat enabling it to float in water. The size of this force is equal to the weight of fluid displaced by the boat.

Figure 2.29 Upthrust U depends on how much water is displaced.



Power and velocity

If $\text{power} = \frac{\text{work done}}{\text{time}}$ then we can

also write $P = \frac{F\Delta s}{t}$ so $P = F\left(\frac{\Delta s}{t}\right)$

which is the same as $P = Fv$ where v is the velocity.

Horse power

Horse power is often used as the unit for power when talking about cars and boats.

$746 \text{ W} = 1 \text{ hp}$

So in Example 1, the power of the car is 105 hp.

to lift all 1000 kg at the same time or move each bag in 0.005 s, both of which are impossible. Power is the quantity that distinguishes between these two tasks.

Power is defined as:

Power = work done per unit time.

Unit of power is the J/s which is the same as the watt (W).

Power is a scalar quantity.

Examples

1. The powerful car

We often use the term power to describe cars. A powerful car is one that can accelerate from 0 to 100 km h^{-1} in a very short time. When a car accelerates, energy is being converted from the chemical energy in the fuel to KE. To have a big acceleration the car must gain KE in a short time, hence be powerful.

2. Power lifter

A power lifter is someone who can lift heavy weights, so shouldn't we say they are strong people rather than powerful? A power lifter certainly is a strong person (if they are good at it) but they are also powerful. This is because they can lift a big weight in a short time.

Worked example

A car of mass 1000 kg accelerates from rest to 100 km h^{-1} in 5 seconds. What is the average power of the car?

Solution

$100 \text{ km h}^{-1} = 28 \text{ m s}^{-1}$.

The gain in KE of the car $= \frac{1}{2}mv^2 = \frac{1}{2} \times 1000 \times 28^2 = 392 \text{ kJ}$

If the car does this in 5 s then

$$\text{power} = \frac{\text{work done}}{\text{time}} = \frac{392}{5} = 78.4 \text{ kW}$$

Exercises

- 47 A weightlifter lifts 200 kg 2 m above the ground in 5 s. Calculate the power of the weightlifter in watts.
- 48 In 25 s a trolley of mass 50 kg runs down a hill. If the difference in height between the top and the bottom of the hill is 50 m, how much power will have been dissipated?
- 49 A car moves along a road at a constant velocity of 20 m s^{-1} . If the resistance force acting against the car is 1000 N, what is the power developed by the engine?

Efficiency

When a box is pulled along the floor, the person pulling has to do work. This work is converted into the KE of the box and some of it is done against friction. Since the result they are trying to achieve is to move the box, the energy transferred to KE could be termed 'useful energy' and the rest is 'wasted'. (The waste energy in this example turns into heat, but we will deal with that in the next chapter). The efficiency tells us how good a system is at turning the input energy into useful work.

Efficiency and power

Since the work is going in and out at the same rate, efficiency can also be expressed as

$$\frac{\text{power out}}{\text{power in}}$$

Efficiency is defined as:

$$\frac{\text{useful work out}}{\text{energy put in}}$$

Efficiency has no units.

Efficiency is a scalar quantity.

It is worth noting here that due to the law of conservation of energy, the efficiency can never be greater than 1.

Worked example

A box of mass 10 kg is pulled along the floor for 2 m by a horizontal force of 50 N. If the frictional force is 20 N, what is the efficiency of the system?

Solution



The work done by the pulling force = force $\times$ distance = $50 \times 2 = 100 \text{ J}$

The unbalanced force on the box = $50 - 20 = 30 \text{ N}$

So the work done on the box = $30 \times 2 = 60 \text{ J}$

This work is exchanged to the box and increases its KE.

Work done against friction = $20 \times 2 = 40 \text{ J}$

So work in = 100 J and total work out = $60 + 40 = 100 \text{ J}$

$$\text{Efficiency} = \frac{\text{useful work out}}{\text{work in}} = \frac{60}{100} = 0.6$$

Expressed as a percentage, this is 60%.

Exercise

- 50** A motor is used to lift a 10 kg mass 2 m above the ground in 4 s. If the power input to the motor is 100 W, what is the efficiency of the motor?
- 51** A motor is 70% efficient. If 60 kJ of energy is put into the engine, how much work is got out?
- 52** The drag force that resists the motion of a car travelling at 80 km h^{-1} is 300 N.
- (a) What power is required to keep the car travelling at that speed?
- (b) If the efficiency of the engine is 60%, what is the power of the engine?

2.10 Uniform circular motion

Assessment statements

- 2.4.1 Draw a vector diagram to illustrate that the acceleration of a particle moving with constant speed in a circle is directed towards the centre of the circle.
- 2.4.2 Apply the expression for centripetal acceleration.
- 2.4.3 Identify the force producing circular motion in various situations.
- 2.4.4 Solve problems involving circular motion.

Circular motion quantities

When describing motion in a circle we often use quantities referring to the angular rather than linear quantities.

Time period (T)

The time taken for one complete circle.

Angular displacement (θ)

The angle swept out by a line joining the body to the centre.

Angular velocity (ω)

The angle swept out per unit time.

$$\omega = \left(\frac{2\pi}{T} \right)$$

Frequency (f)

The number of complete circles per unit time ($f = \frac{1}{T}$).

Size of centripetal acceleration

The centripetal acceleration is given by the following formula

$$a = \frac{v^2}{r}$$

Or in circular terms

$$a = \omega^2 r$$

If a car travels around a bend at 30 km h^{-1} , it is obviously travelling at a constant speed, since the speedometer would register 30 km/hr all the way round. However it is not travelling at constant velocity. This is because velocity is a vector quantity and for a vector quantity to be constant, both magnitude and direction must remain the same. Bends in a road can be many different shapes, but to simplify things, we will only consider circular bends.

Centripetal acceleration

From the definition of acceleration, we know that if the velocity of a body changes, it must be accelerating, and that the direction of acceleration is in the direction of the change in velocity. Let us consider a body moving in a circle with a constant speed v . Figure 2.86 shows two positions of the body separated by a short time.

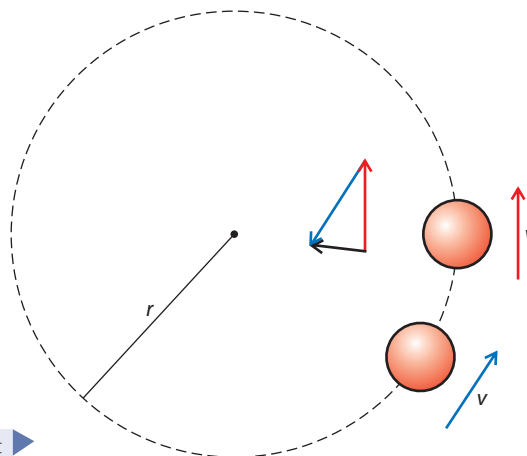


Figure 2.86 A body travels at constant speed around a circle of radius r .

Since this is a two-dimensional problem, we will need to draw the vectors to find out the change in velocity. This has been done on the diagram. It can be seen that the vector representing the change in velocity points to the centre of the circle. This means that the acceleration is also directed towards the centre, and this acceleration is called *centripetal acceleration*. All bodies moving in a circle accelerate towards the centre.

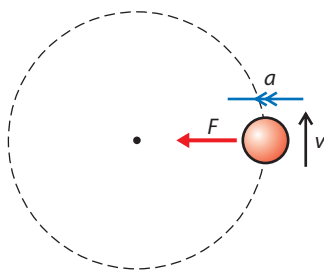


Figure 2.87 Showing the direction of the force and acceleration.

Centripetal force

From Newton's first law, we know that if a body accelerates, there must be an unbalanced force acting on it. The second law tells us that that force is in the direction of the acceleration. This implies that there must be a force acting towards the centre. This force is called the *centripetal force*.

$$\text{Centripetal force, } F = \frac{mv^2}{r} = m\omega^2 r$$

Examples

All bodies moving in a circle must be acted upon by a force towards the centre of the circle. However, this can be provided by many different forces.

1. Ball on a string

You can make a ball move in a circle by attaching it to a string and swinging it round your head. In this case the centripetal force is provided by the tension in the string. If the string suddenly broke, the ball would fly off in a straight line at a tangent to the circle.

2. Car going round a bend

When a car goes round a bend, the force causing it to move in a circle is the friction between the road and the tyres. Without this friction the car would move in a straight line.

3. Wall of Death

In the Wall of Death the rider travels around the inside walls of a cylinder. Here the centripetal force is provided by the normal force between the wall and the tyres.

Wall of Death



Exercise

- 53** Calculate the centripetal force for a 1000 kg car travelling around a circular track of radius 50 m at 30 km h^{-1} .
- 54** A 200 g ball is made to travel in a circle of radius 1 m on the end of a string. If the maximum force that the string can withstand before breaking is 50 N, what is the maximum speed of the ball?
- 55** A body of mass 250 g moves in a circle of radius 50 cm with a speed of 2 m s^{-1} . Calculate
- the distance travelled per revolution
 - the time taken for one revolution
 - the angle swept out per revolution
 - the angular velocity
 - the centripetal acceleration
 - the centripetal force.



Centripetal is not an extra force

Remember when solving circular motion problems, centripetal force is not an extra force – it is one of the existing forces. Your task is to find which force (or a component of it) points towards the centre.

Practice questions

- 1** This question is about linear motion.
- A police car P is stationary by the side of a road. A car S, exceeding the speed limit, passes the police car P at a constant speed of 18 m s^{-1} . The police car P sets off to catch car S just as car S passes the police car P. Car P accelerates at 4.5 m s^{-2} for a time of 6.0 s and then continues at constant speed. Car P takes a time t seconds to draw level with car S.
- State an expression, in terms of t , for the distance car S travels in t seconds. (1)
 - Calculate the distance travelled by the police car P during the first 6.0 seconds of its motion. (1)
 - Calculate the speed of the police car P after it has completed its acceleration. (1)
 - State an expression, in terms of t , for the distance travelled by the police car P during the time that it is travelling at constant speed. (1)
 - Using your answers to (a), determine the total time t taken for the police car P to draw level with car S. (2)

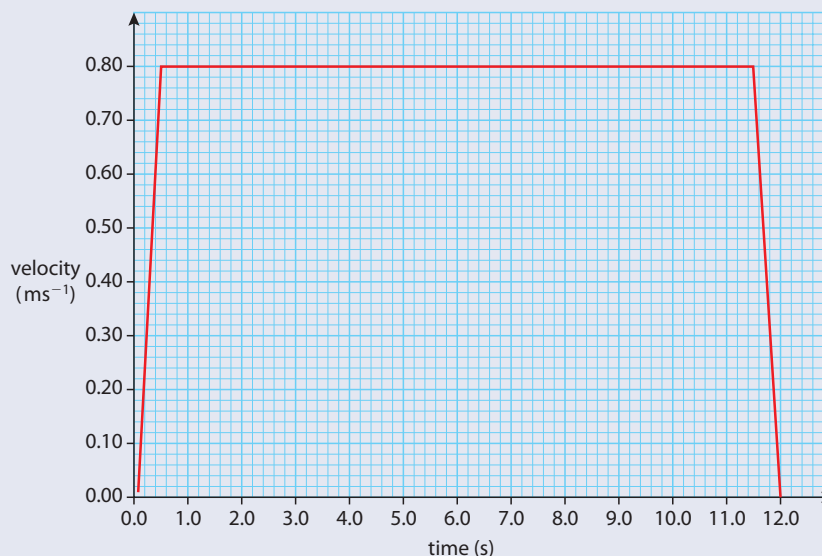
(Total 6 marks)

© International Baccalaureate Organisation

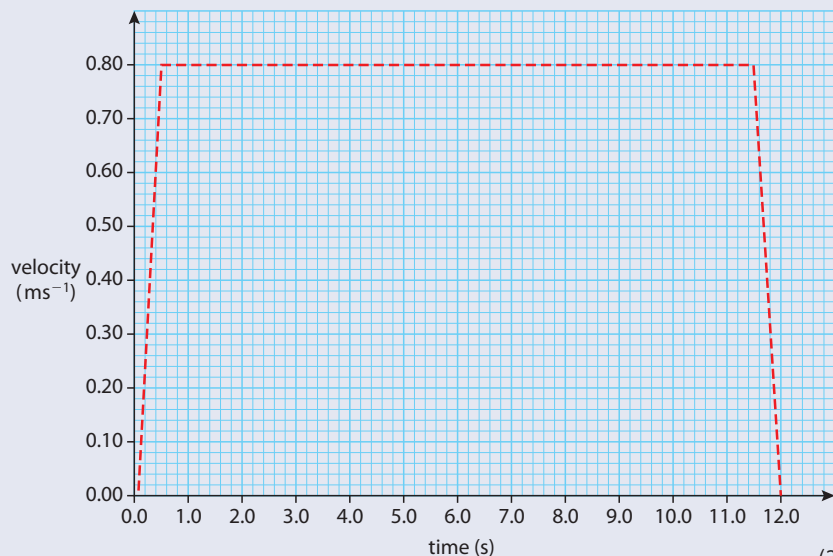
2 This question is about the kinematics of an elevator (lift).

- (a) Explain the difference between the gravitational mass and the inertial mass of an object. (3)

An elevator (lift) starts from rest on the ground floor and comes to rest at a higher floor. Its motion is controlled by an electric motor. A simplified graph of the variation of the elevator's velocity with time is shown below.

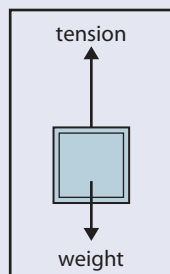


- (b) The mass of the elevator is 250 kg. Use this information to calculate
- (i) the acceleration of the elevator during the first 0.50 s. (2)
 - (ii) the total distance travelled by the elevator. (2)
 - (iii) the minimum work required to raise the elevator to the higher floor. (2)
 - (iv) the minimum average power required to raise the elevator to the higher floor. (2)
 - (v) the efficiency of the electric motor that lifts the elevator, given that the input power to the motor is 5.0 kW. (2)
- (c) On the graph axes below, sketch a realistic variation of velocity for the elevator. Explain your reasoning. (*The simplified version is shown as a dotted line*)



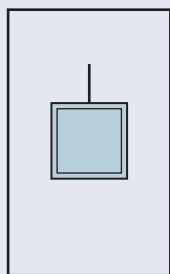


The elevator is supported by a cable. The diagram opposite is a free-body force diagram for when the elevator is moving upwards during the first 0.50 s.

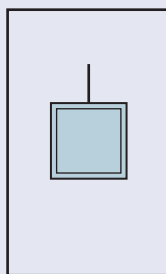


(d) In the space below, draw free-body force diagrams for the elevator during the following time intervals.

(i) 0.5 to 11.50 s



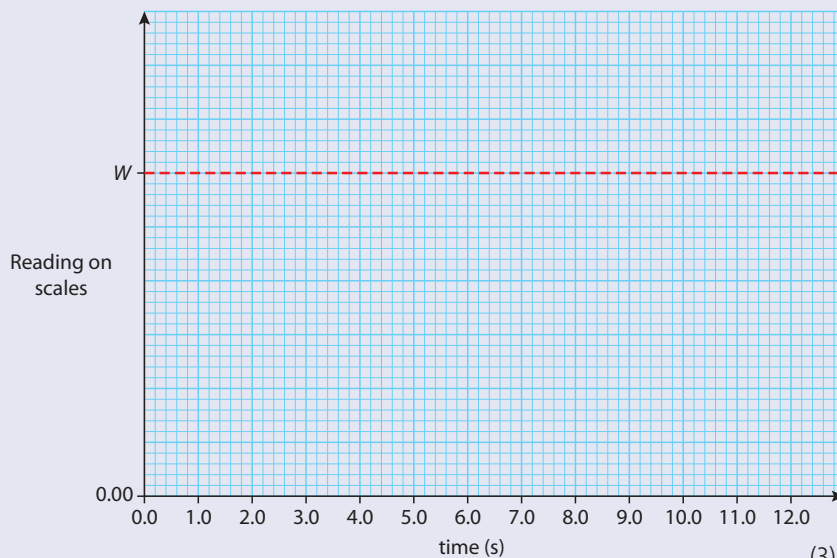
(ii) 11.50 to 12.00 s



(3)

A person is standing on weighing scales in the elevator. Before the elevator rises, the reading on the scales is W .

(e) On the axes below, sketch a graph to show how the reading on the scales varies during the whole 12.00 s upward journey of the elevator. (*Note that this is a sketch graph – you do not need to add any values.*)



(3)

(f) The elevator now returns to the ground floor where it comes to rest. Describe and explain the energy changes that take place during the whole up and down journey.

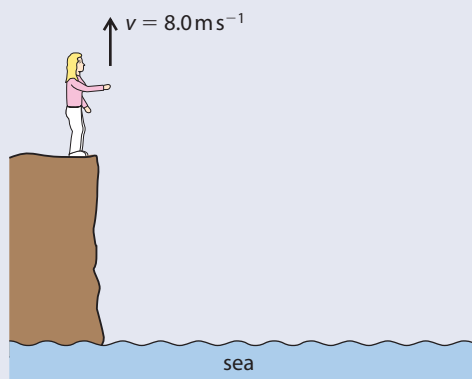
(4)

(Total 25 marks)

© International Baccalaureate Organisation



- 3 This question is about throwing a stone from a cliff.
Antonia stands at the edge of a vertical cliff and throws a stone vertically upwards.



The stone leaves Antonia's hand with a speed $v = 8.0 \text{ m s}^{-1}$.

The acceleration of free fall g is 10 m s^{-2} and all distance measurements are taken from the point where the stone leaves Antonia's hand.

- (a) Ignoring air resistance calculate
- (i) the maximum height reached by the stone. (2)
 - (ii) the time taken by the stone to reach its maximum height. (1)

The time between the stone leaving Antonia's hand and hitting the sea is 3.0 s.

- (b) Determine the height of the cliff. (3)

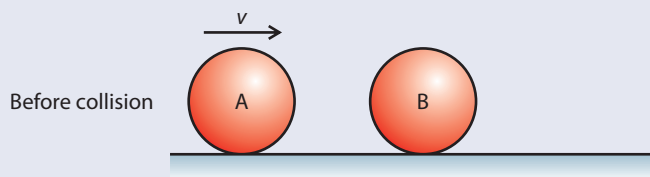
(Total 6 marks)

© International Baccalaureate Organisation

- 4 This question is about conservation of momentum and conservation of energy.

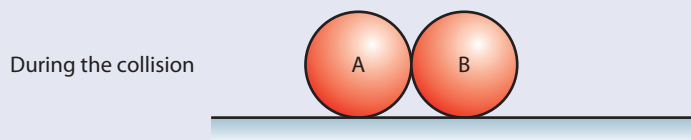
- (a) State Newton's third law. (1)
- (b) State the law of conservation of momentum. (2)

The diagram below shows two identical balls A and B on a horizontal surface. Ball B is at rest and ball A is moving with speed v along a line joining the centres of the balls. The mass of each ball is M .



During the collision of the balls, the magnitude of the force that ball A exerts on ball B is F_{AB} and the magnitude of the force that ball B exerts on ball A is F_{BA} .

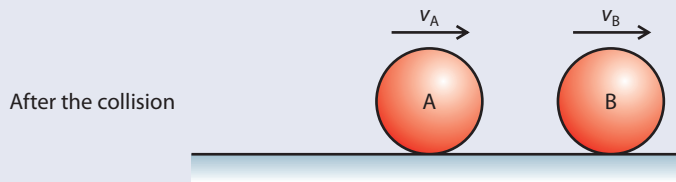
- (c) On the diagram below, add labelled arrows to represent the magnitude and direction of the forces F_{AB} and F_{BA} .



(3)



The balls are in contact for a time t . After the collision, the speed of ball A is $+v_A$ and the speed of ball B is $+v_B$ in the directions shown.



As a result of the collision, there is a change in momentum of ball A and of ball B.

- (d) Use Newton's second law of motion to deduce an expression relating the forces acting during the collision to the change in momentum of
- (i) ball B. (2)
 - (ii) ball A. (2)
- (e) Apply Newton's third law and your answers to (d) to deduce that the change in momentum of the system (ball A and ball B) as a result of this collision, is zero. (4)
- (f) Deduce that if kinetic energy is conserved in the collision, then after the collision, ball A will come to rest and ball B will move with speed v . (3)

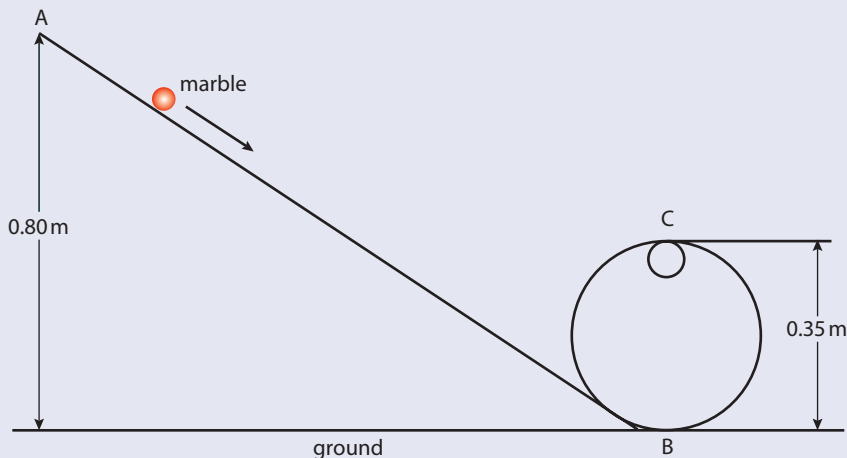
(Total 17 marks)

© International Baccalaureate Organisation

5 This question is about the kinematics and dynamics of circular motion.

- (a) A car goes round a curve in a road at constant speed. Explain why, although its speed is constant, it is accelerating. (2)

In the diagram below, a marble (small glass sphere) rolls down a track, the bottom part of which has been bent into a loop. The end A of the track, from which the marble is released, is at a height of 0.80 m above the ground. Point B is the lowest point and point C the highest point of the loop. The diameter of the loop is 0.35 m.



The mass of the marble is 0.050 kg. Friction forces and any gain in kinetic energy due to the rotating of the marble can be ignored. The acceleration due to gravity, $g = 10 \text{ m s}^{-2}$.

Consider the marble when it is at point C.

- (b) (i) On the diagram, draw an arrow to show the direction of the resultant force acting on the marble. (1)
- (ii) State the names of the two forces acting on the marble. (2)
- (iii) Deduce that the speed of the marble is 3.0 m s^{-1} . (3)



- (iv) Determine the resultant force acting on the marble and hence determine the reaction force of the track on the marble. (4)

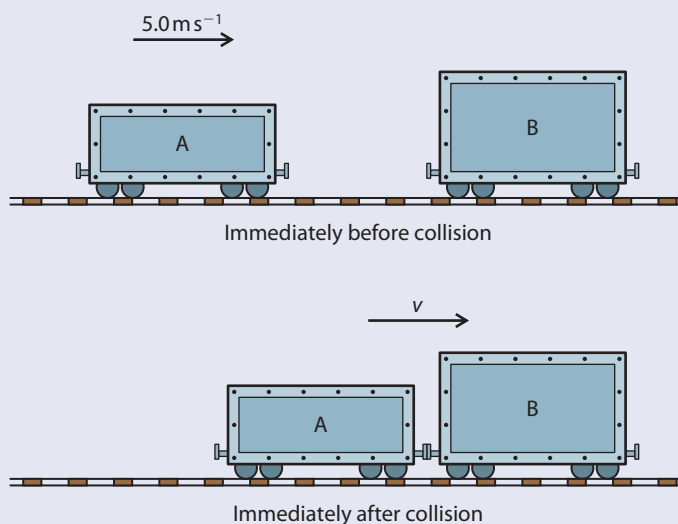
(Total 12 marks)

© International Baccalaureate Organisation

- 6 This question is about the collision between two railway trucks (carts).

(a) Define *linear momentum*. (1)

In the diagram below, railway truck A is moving along a horizontal track. It collides with a stationary truck B and on collision, the two join together. Immediately before the collision, truck A is moving with speed 5.0 m s^{-1} . Immediately after collision, the speed of the trucks is v .



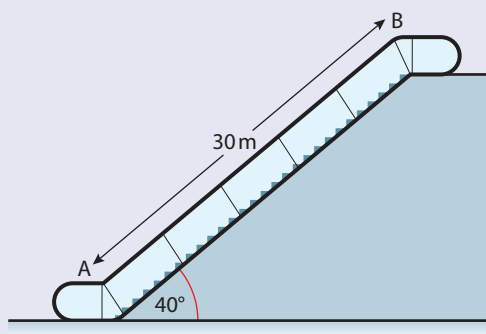
The mass of truck A is 800 kg and the mass of truck B is 1200 kg .

- (b) (i) Calculate the speed v immediately after the collision. (3)
 (ii) Calculate the total kinetic energy lost during the collision. (2)
 (c) Suggest what has happened to the lost kinetic energy. (2)

(Total 8 marks)

© International Baccalaureate Organisation

- 7 This question is about estimating the energy changes for an escalator (moving staircase). The diagram below represents an escalator. People step on to it at point A and step off at point B.





4

Simple harmonic motion and waves

4.1

Kinematics of simple harmonic motion

Assessment statements

- 4.1.1 Describe examples of oscillations.
- 4.1.2 Define the terms *displacement*, *amplitude*, *frequency*, *period* and *phase difference*.
- 4.1.3 Define *simple harmonic motion* (SHM) and state the defining equation as $a = -\omega^2 x$.
- 4.1.4 Solve problems using the defining equation for SHM.
- 4.1.5 Apply the equations $v = v_0 \sin \omega t$, $v = v_0 \cos \omega t$, $v = \pm \omega \sqrt{(x_0^2 - x^2)}$, $x = x_0 \cos \omega t$ and $x = x_0 \sin \omega t$ as solutions to the defining equation for SHM.
- 4.1.6 Solve problems, both graphically and by calculation, for acceleration, velocity and displacement during SHM.

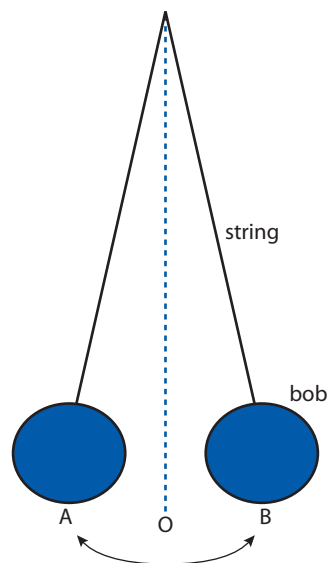
A swing is an example of oscillatory motion.



Oscillations

In this section we will derive a mathematical model for an oscillating or vibrating body. There are many different examples of naturally occurring oscillations but they don't all have the same type of motion. We are going to consider the simplest form of oscillation: simple harmonic motion. The most common example of this is a pendulum (Figure 4.1). Before we start to model this motion, we need to define some new terms and quantities.

Figure 4.1 The simple pendulum swings from A to B and back.





Cycle

One cycle is defined as one complete oscillation of the pendulum (A–B–A). The term *cycle* is also used to describe circular motion; one cycle is one complete circle or 2π radians.

Equilibrium position

The equilibrium position is the position where the pendulum bob would rest if not disturbed – this is position O.

Amplitude (x_0)

The amplitude is defined as the maximum displacement from the equilibrium position, this is distance OB or OA.

Unit: metre

Time period (T)

The time period is the time taken for one complete cycle.

Unit: second

Frequency (f)

The frequency is the number of cycles that the pendulum makes per unit time. This is equal to $1/\text{time period}$.

Unit: s^{-1} or hertz (Hz)

Angular frequency (ω)

The angular frequency is found by multiplying f by 2π ($\omega = 2\pi f$). This quantity is normally used when describing circular motion. An angular frequency of $2\pi \text{ rad s}^{-1}$ means that a body makes one revolution per second. However, it is also used to describe an oscillation, 2π being equivalent to one complete cycle.

Unit: s^{-1} or hertz (Hz)

Worked example

A pendulum completes 10 swings in 8 s. What is the angular frequency?

Solution

There are 10 swings in 8 seconds, so each swing takes 0.8 s.

Time period = 0.8 s.

$$\text{Frequency} = \frac{1}{T} = \frac{1}{0.8} = 1.25 \text{ Hz}$$

$$\text{Angular frequency } \omega = 2\pi f = 2\pi \times 1.25 = 7.8 \text{ rad s}^{-1}$$



Measuring time period

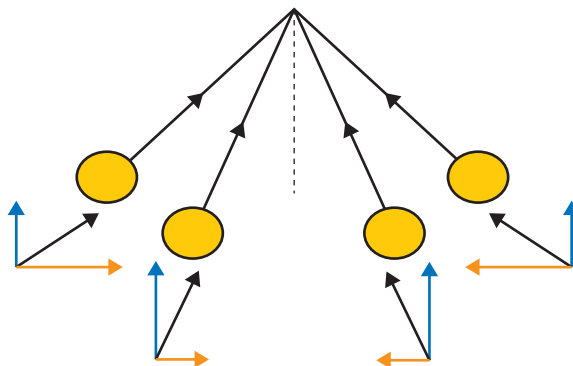
To reduce the uncertainties when measuring the time period of an oscillation, it is best to measure the time for many oscillations e.g. 10. The time period is found by dividing this time by the number of oscillations.

Analysing oscillations

To make a model of oscillatory motion, we will analyse two different oscillations and see if there are any similarities.

The pendulum

Figure 4.2 As the angle increases, the horizontal component of tension increases, but is always pointing towards the centre.

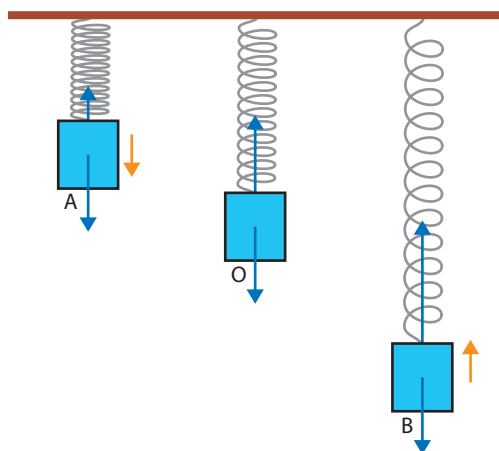


When a pendulum bob is pushed to one side and released, it will swing back down. The reason for this can be understood by drawing the forces acting on the bob. In Figure 4.2, we can see that when the string makes an angle to the vertical, the tension has a component in the horizontal direction; this component causes the bob to accelerate back towards the middle. As the bob swings down, the angle of the string gets smaller, and the horizontal component decreases. The horizontal acceleration of the bob is proportional to the horizontal force, so we can therefore deduce that the horizontal acceleration is proportional to the displacement from the lowest point.

When the ball reaches the lowest position, it is travelling at its maximum speed. It passes through this position and continues to swing up on the other side. The horizontal component of the tension is now acting in the other direction. This is in the *opposite* direction to the motion so will slow the bob down. We can conclude that no matter where the bob is, its acceleration is always directed towards O.

Mass on a spring

Figure 4.3 The tension increases as the spring is stretched. The resultant (red) increases with increased distance from the centre and is always directed towards the centre.



If a mass hanging on the end of a spring is lifted up and released, it will bounce up and down as in Figure 4.3. The forces acting on the mass are *weight* and the *tension in the spring*. The weight is always the same but the tension depends on how far the spring is stretched. (When you stretch a spring, the tension is proportional to the extension).



At A, the spring is short, so the tension will be small; the weight will therefore be bigger than the tension, so the resultant force will be downwards.

As the mass passes through the middle point, the forces will be balanced.

At B, the spring is stretched, so the tension is large; the tension will therefore be greater than the weight, so the resultant force will be upwards.

Again we can see that the acceleration is proportional to the displacement from the central point and always directed towards it.

This type of motion is called *simple harmonic motion* or SHM.

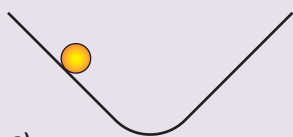


SHM

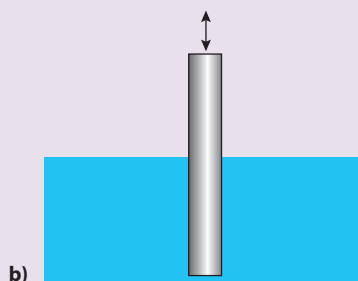
The acceleration is proportional to the distance from a fixed point.
The acceleration is always directed towards a fixed point.

Exercises

- 1 State whether the following are examples of simple harmonic motion.



a)



b)

Figure 4.4

- (a) A ball rolling up and down on a track (Figure 4.4a).
(b) A cylindrical tube floating in water when pushed down and released (Figure 4.4b).
(c) A tennis ball bouncing back and forth across the net.
(d) A bouncing ball.
- 2 A pendulum completes 20 swings in 12 s. What is
(a) the frequency?
(b) the angular frequency?

Graphical treatment

To analyse the oscillation further, we can plot graphs for the motion. In this example, we will consider a mass on a spring, but we could choose any simple harmonic motion.

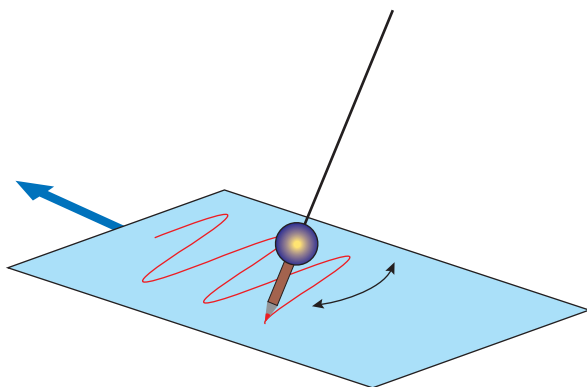


Figure 4.5 You can plot a displacement–time graph by attaching a pen to a pendulum and moving paper beneath it at a constant velocity.



To see how the equation fits the graph we can put some numbers into the equation.

In this example, the time period = 4 s

Therefore $f = \frac{1}{4} = 0.25 \text{ Hz}$

Angular frequency = $2\pi f = 0.5\pi$

So displacement = $2 \cos(0.5\pi t)$

Calculating displacement at different times gives:

$t = 1 \text{ s } y = 2 \cos(\pi/2) = 0 \text{ cm}$

$t = 2 \text{ s } y = 2 \cos(\pi) = -2 \text{ cm}$

$t = 3 \text{ s } y = 2 \cos(3\pi/2) = 0 \text{ cm}$

$t = 4 \text{ s } y = 2 \cos(2\pi) = 2 \text{ cm}$



Displacement–time

As before, O is the equilibrium position and we will take this to be our position of zero displacement. Above this is positive displacement and below is negative.

At A, the mass has maximum positive displacement from O.

At O, the mass has zero displacement from O.

At B, the mass has maximum negative displacement from O.

We can see that the shape of this displacement–time graph is a cosine curve.

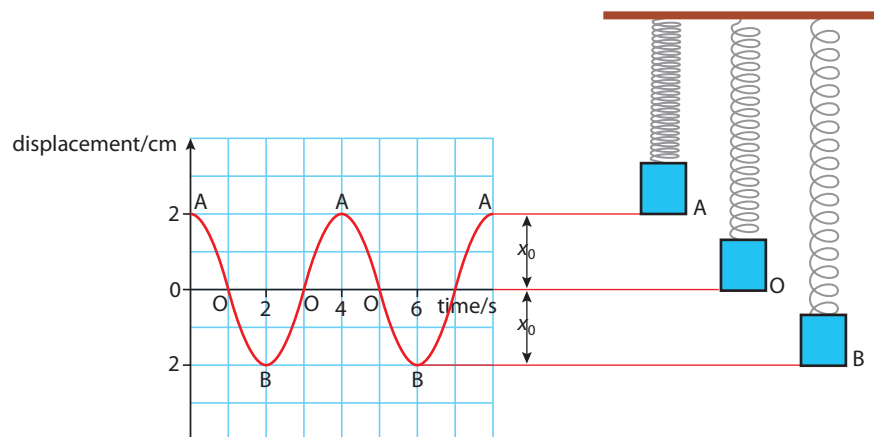


Figure 4.6 Displacement–time graph.

The equation of this line is $x = x_0 \cos \omega t$,

where x_0 is the maximum displacement and ω is the angular frequency.

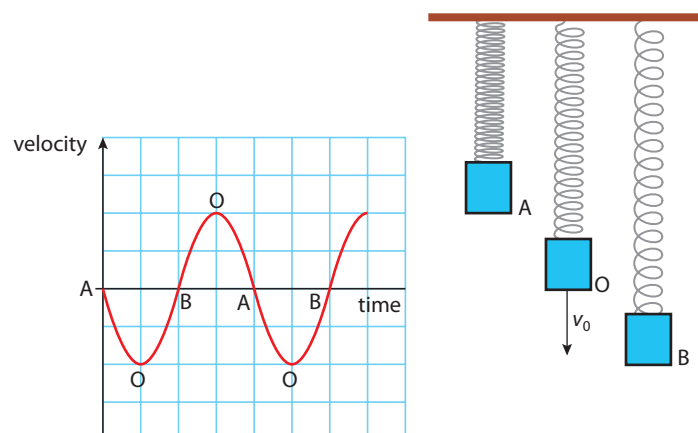
Velocity–time

From the gradient of the displacement–time graph (Figure 4.6), we can calculate the velocity.

At A, gradient = 0 so velocity is zero.

At O, gradient is negative and maximum, so velocity is down and maximum.

At B, gradient = 0 so velocity is zero.



The equation of this line is $v = -v_0 \sin \omega t$ where v_0 is the maximum velocity.

Acceleration–time

From the gradient of the velocity–time graph (Figure 4.7) we can calculate the acceleration.

At A, the gradient is maximum and negative so acceleration is maximum and downwards.

At O, the gradient is zero so acceleration is zero.

At B, the gradient is maximum and positive so the acceleration is maximum and upwards.

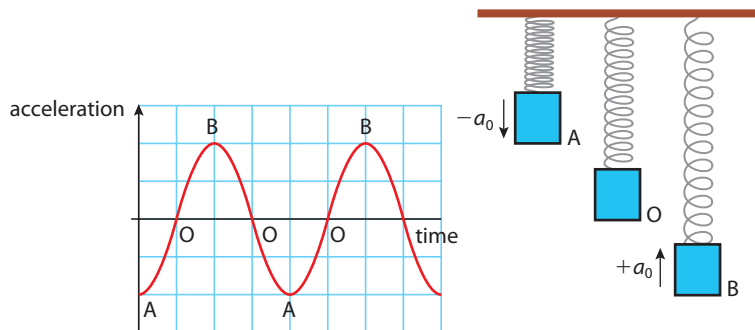


Figure 4.8 Acceleration–time graph.

The equation of this line is $a = -a_0 \cos \omega t$ where a_0 is this maximum acceleration.

So $x = x_0 \cos \omega t$ and $a = -a_0 \cos \omega t$

When displacement increases, acceleration increases proportionally but in a negative sense; in other words: $a \propto -x$

We have confirmed that the acceleration of the body is directly proportional to the displacement of the body and always directed towards a fixed point.

Worked example

A mass on a spring is oscillating with a frequency 0.2 Hz and amplitude 3.0 cm. What is the displacement of the mass 10.66 s after it is released from the top?

Solution

$$x = x_0 \cos \omega t. \quad \text{Since this is SHM}$$

where x = displacement

$$x_0 = \text{amplitude} = 3 \text{ cm}$$

$$\begin{aligned} \omega &= \text{angular velocity} = 2\pi f = 2\pi \times 0.2 \\ &= 0.4\pi \text{ Hz} \end{aligned}$$

$$t = \text{time} = 10.66 \text{ s}$$

$$x = 0.03 \times \cos(0.4\pi \times 10.66) \quad \text{Substituting into the formula}$$

$$x = 0.02 \text{ m}$$

$$= 2 \text{ cm}$$



Radians

When calculating $\cos \omega t$, you must have your calculator set on radians.

Exercises

- 3 For the same mass on a spring in the example above, calculate the displacement after 1.55 s.
- 4 Draw a displacement time sketch graph for this motion.
- 5 A long pendulum has time period 10 s. If the bob is displaced 2 m from the equilibrium position and released, how long will it take to move 1 m?
- 6 As a mass on a spring travels upwards through the equilibrium position, its velocity is 0.5 m s^{-1} . If the frequency of the pendulum is 1 Hz what will the velocity of the bob be after 0.5 s?

Figure 4.9 A short time after the ball starts moving, the radius makes an angle θ with the horizontal.

Figure 4.10 When a ball moving in a circle is viewed from the side, it looks like it is moving with SHM.

Figure 4.11 From the triangle we can see that the horizontal displacement $x = x_0 \cos \theta$.

Figure 4.12 Horizontal velocity vectors.

$$\begin{aligned}\text{Speed, } v &= \frac{\text{distance}}{\text{time}} \\ &= \frac{\text{circumference}}{\text{time period}} \\ &= \frac{2\pi r}{T}\end{aligned}$$

$$\text{But } \frac{2\pi}{T} = \omega$$

$$\text{So speed} = \omega r$$

$$\begin{aligned}\text{Centripetal acceleration} &= \frac{v^2}{r} \\ &= \frac{\omega^2 r^2}{r} \\ &= \omega^2 r\end{aligned}$$

Figure 4.13 Horizontal acceleration vectors.

If you have done differentiation in maths then you will understand that if

$$\text{displacement } x = x_0 \cos \omega t$$

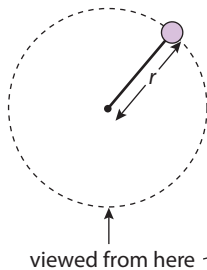
$$\text{then velocity } \frac{dx}{dt} = -x_0 \omega \sin \omega t$$

$$\text{and acceleration, } \frac{d^2x}{dt^2} = -x_0 \omega^2 \cos \omega t$$

$$\text{This implies that } a = -\omega^2 x$$

This is a much shorter way of deriving this result!

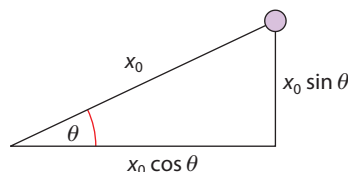
SHM and circular motion



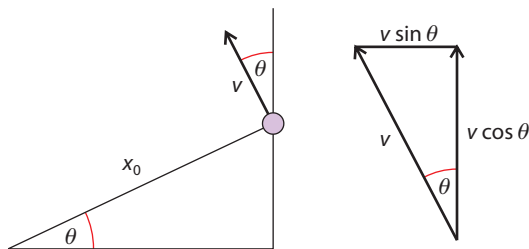
If we analyse the motion of the ball in Figure 4.9, we find that it is also SHM. The ball is travelling in a circle of radius x_0 with a constant speed v . The ball takes a time T to complete one revolution.

Let us consider the horizontal component of motion. We can write equations for this component.

Displacement



Velocity



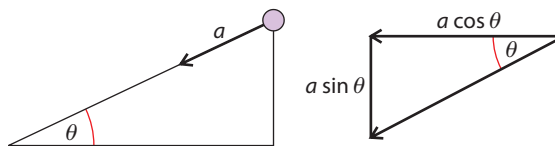
The horizontal velocity $= -v \sin \theta$

But for circular motion $v = \omega r$ so in this case $v = \omega x_0$

So horizontal velocity $= -\omega x_0 \sin \theta$

Acceleration

When bodies travel in a circle, they have an acceleration towards the centre (the centripetal acceleration) $a = \omega^2 r$. In this case, acceleration is $\omega^2 x_0$ since the radius is x_0 .



Horizontal component of acceleration $= -a \cos \theta$

But $a = \omega^2 x_0$

So horizontal acceleration $= -\omega^2 x_0 \cos \theta$

Now we have found that the displacement $x = x_0 \cos \theta$

So acceleration $= -\omega^2 x$

So the horizontal acceleration is proportional to the displacement, and is always directed towards the centre. In other words, the horizontal component of the motion is SHM. We have also found out that the constant of proportionality is ω^2 .

Now we have concluded that this motion is SHM we can use the equations that we have derived to model all simple harmonic motions.

Equations for SHM

$$\text{Displacement} = x_0 \cos \omega t \quad (1)$$

$$\text{Velocity} = -\omega x_0 \sin \omega t \quad (2)$$

$$\text{Acceleration} = -\omega^2 x_0 \cos \omega t \quad (3)$$

$$\text{We also know that } a = -\omega^2 x$$

$$\text{From Pythagoras, } 1 = \sin^2 \theta + \cos^2 \theta$$

$$\text{So, } \sin \theta = \sqrt{1 - \cos^2 \theta} \quad \text{Rearranging}$$

$$\text{Therefore } \sin \omega t = \sqrt{1 - \cos^2 \omega t} \quad \text{Substituting for } \theta = \omega t$$

$$\text{Multiplying by } \omega \text{ gives } \omega x_0 \sin \omega t = \omega x_0 \sqrt{1 - \cos^2 \omega t}$$

$$\omega x_0 \sin \omega t = \omega \sqrt{x_0^2 - x_0^2 \cos^2 \omega t} \quad \text{Taking } x_0 \text{ into the square root}$$

$$\text{But from equation (1) } x_0^2 \cos^2 \omega t = x^2$$

$$\text{So } v = \omega \sqrt{x_0^2 - x^2} \quad \text{Substituting}$$

The maximum velocity is when the displacement is 0 so $x = 0$

$$\text{Maximum velocity} = \omega x$$

Worked example

- 1 A pendulum is swinging with a frequency of 0.5 Hz. What is the size and direction of the acceleration when the pendulum has a displacement of 2 cm to the right?
- 2 A pendulum bob is swinging with SHM at a frequency of 1 Hz and amplitude 3 cm. At what position will the bob be moving with maximum velocity and what is the size of the velocity?

Solution

- 1 Assuming the pendulum is swinging with SHM, then we can use the equation $a = \omega^2 x$ to calculate the acceleration.

$$\omega = 2\pi f = 2\pi \times 0.5 = \pi$$

$$a = -\pi^2 \times 0.02 = -0.197 \text{ m s}^{-2} \quad \text{Since -ve direction is to the left}$$

- 2 $v = \omega \sqrt{x_0^2 - x^2}$ Since the motion is SHM

This is maximum when $x = 0$ This is when the pendulum swings through the central position

$$\text{The maximum value} = \omega x_0 \text{ where } \omega = 2\pi f = 2 \times \pi \times 1 = 2\pi \text{ rad s}^{-1}$$

$$\text{Maximum } v = 2\pi \times 0.03 = 0.188 \text{ m s}^{-1}$$

Summary

If a body oscillating with SHM has an angular frequency $\omega = 2\pi f$ and amplitude x_0 then its displacement (x), velocity (v) and acceleration (a) at any given time can be found from the following equations:

$$x = x_0 \cos \omega t$$

$$v = -x_0 \omega \sin \omega t$$

$$a = -\omega^2 x_0 \cos \omega t$$

In addition, at a given displacement x , the velocity and acceleration can be found from the following equations:

$$v = \omega \sqrt{x_0^2 - x^2}$$

$$\text{Maximum velocity} = \omega x_0$$

$$a = -\omega^2 x$$

The real pendulum

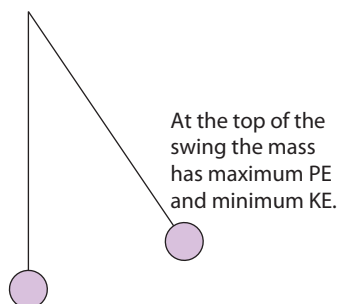
The pendulum is a classic example of SHM. However it is only SHM if the swings are very small (less than 10°). This is worth remembering if you ever do an experiment with a real pendulum.

Exercises

- 7 A long pendulum swings with a time period of 5 s and an amplitude of 2 m.
 - (a) What is the maximum velocity of the pendulum?
 - (b) What is the maximum acceleration of the pendulum?
- 8 A mass on a spring oscillates with amplitude 5 cm and frequency 2 Hz. The mass is released from its highest point. Calculate the velocity of the mass after it has travelled 1 cm.
- 9 A body oscillates with SHM of time period 2 s. What is the amplitude of the oscillation if its velocity is 1 m s^{-1} as it passes through the equilibrium position?

4.2

Energy changes during simple harmonic motion (SHM)



At the bottom of the swing the mass has maximum KE and minimum PE.

Figure 4.14 In the simple pendulum, energy is changing from one form to another as it moves.



To view the PhET *Masses and springs* simulation, visit heinemann.co.uk/hotlinks, enter the express code 4426P and click on Weblink 4.1.

Assessment statements

- 4.2.1 Describe the interchange between kinetic energy and potential energy during SHM.
- 4.2.2 Apply the expression $E_k = \frac{1}{2}m\omega^2(x_0^2 - x^2)$ for the kinetic energy of a particle undergoing SHM, $E_T = \frac{1}{2}m\omega^2x_0^2$ for the total energy and $E_p = \frac{1}{2}m\omega^2x^2$ for the potential energy.
- 4.2.3 Solve problems, both graphically and by calculation, involving energy changes during SHM.

If we once again consider the simple pendulum, we can see that its energy changes as it swings.

Kinetic energy

We have already shown that the velocity of the mass is given by the equation

$$v = \omega\sqrt{x_0^2 - x^2}$$

From definition, $KE = \frac{1}{2}mv^2$

Substituting: $KE = \frac{1}{2}m\omega^2(x_0^2 - x^2)$

KE is a maximum at the bottom of the swing where $x = 0$.

So $KE_{\max} = \frac{1}{2}m\omega^2x_0^2$

At this point the PE is zero.

Total energy

The total energy at any moment in time is given by:

$$\text{total energy} = KE + PE$$

So at the bottom of the swing:

$$\text{total energy} = \frac{1}{2}m\omega^2x_0^2 + 0 = \frac{1}{2}m\omega^2x_0^2$$

Since no work is done on the system, according to the law of conservation of energy, the total energy must be constant.

$$\text{So total energy} = \frac{1}{2}m\omega^2x_0^2$$

Potential energy

Potential energy at any moment = total energy – KE

$$\text{So } PE = \frac{1}{2}m\omega^2x_0^2 - \frac{1}{2}m\omega^2(x_0^2 - x^2)$$

$$PE = \frac{1}{2}m\omega^2x^2$$